

第4回数理構造セミナー

日時：2018年4月12日（木） 11:00～18:00

場所：すずかけ台キャンパス、S1棟

プログラム：

11:00-11:30 (S1-203) Hayato Goto: Modeling Evolution of Business Network and Their Interaction

11:30-14:00 (S1-second floor) Poster session with lunch (S1-203)

14:00-15:15 (S1-317-2) Murat Erkurt: "Emergence of Form in Embryogenesis"

15:30-16:45 (S1-317-2) Mauricio Barahona: "Graph partitioning and community detection based on dynamics"

16:45-18:00 (S1-second floor) Poster session with beer (S1-203)

Murat Erkurt: Abstract:

Development of form in an embryo is the result of a series of symmetry breakings starting from topological and informational singularity of the zygote. Pattern generating reaction-diffusion systems are not capable of form generation, as they are locally isotropic. We introduce Vector-Reaction-Diffusion-Drift system based on three new concepts: source-field coupling, local drift for global symmetry breaking and vector morphogens as pillars of form. An integrated model with topological, physical, chemical and regulatory layers is presented. Simulation of blastula moving into gastrulation shows realistic form generation as an emergent phenomenon of the ensemble of locally interacting identical cells. We further expand our model by introducing Dynamical Cell Differentiation where the regulatory genome is generalized into a finite-state-machine. This allows symmetry breakings to be choreographed into sub-spatial domains resulting in further detailed form generation.

Posters (to be posted until 20th):

●Arthur Matsuo: "Power-Law Distributions from Sigma-Pi Structure of Sums of Random Multiplicative Processes"

●Emiko Takagi: "Category classification of Information diffusion in Blogosphere network based on degree of similarity between documents"

●Gen Sakoda: "Parameter estimation method for non-stationary Poisson time series with fluctuation scaling and incomplete observation"

●Tomoya Hamano : "Dealer model of foreign exchange market with finite assets"

●Takumi Sueshige: "Ecology of Forex market: Human vs. Artificial Intelligence"

●Yuh Kobayashi: "Evolutionary Flow and Scaling Laws of Business Firms in 3-dimensional State Space"

●Jean Francois Boilard: "Market Microstructure Analysis of Limit Orders in FX Market"

●Yohei Shida: "Deriving mass human mobility patterns from individuals location data"

Kiyoshi Kanazawa: "Boltzmann Equation for Financial Brownian Motion"

●Junichi Ozaki:?

●Fernando Rosas:?

●Takayuki Nozawa, Mutsumi Kondo, Reiko Yamamoto, Hyeonjeong Jeong, Shigeyuki Ikeda, Kohei Sakaki, Yasushige Ishikawa, Yoshihiro Miyake,

●Ryuta Kawashima: " Shared subjective experience and interpersonal neural synchronization in foreign language active learning classroom: a pilot study"

●Ken-ichiro Ogawa, Yoshihiro Miyake: " Robust patterning of gene expression based on internal coordinate system of cells "

●Hiroki Ora, Yoshihiro Miyake: " A method for prediction of behavioral errors from single-trial electrophysiological data"

●Shun-ichi Amano, Ken-ichiro Ogawa, Yoshihiro Miyake: "Node property of weighted networks considering connectability to nodes within two degrees of separation"

●Taisei Mukai, Takao Terano: "Effects of Trade Environment in the Decentralized Inter-firm trade Structure Model through Agent Simulation"

第 5 回数理構造セミナー

日時 : 2018 年 4 月 19 日 (木) 14:00~17:30

場所 : すずかけ台キャンパス、S1 棟 3 階 317-2 号室

プログラム :

14:00-15:30 Henrik J. Jensen: Analysis of EEG response from musicians and audience during performances: improvisation versus non-improvisation.

15:45-16:45 Tim Evans: "Networks and Space-time"

16:45-17:30 Paul Expert: "Spectral signature of artificial and real dynamical systems"